

Jeremy Kalfus

jeremykalfus.github.io



EDUCATION

- **Indian Springs School** Pelham, Alabama
High School Diploma Aug. 2022 – Apr. 2026
 - **SAT:** 1570/1600 (780 RW, 790 M)
 - **Extracurricular Involvement:** Scholar's Bowl Team Captain; "Woodward Post" (School Newspaper) Head Web Editor, Writer, and Editor; Investment Club Founder and President; "Jew Crew" President; HOSA Officer; Science Olympiad Member
 - **AP (Advanced Placement) Course-load:** Total APs (4y): 15

EXPERIENCE

- **Apex Neuroscience** Remote
Research Intern, Brain Ischemia & Perfusion Feb. 2026 – Aug. 2026
 - **Focus:** Pre-registered (OSF) systematic review of how long the brain stays perfusable after global cerebral ischemia, with Dr. Andrew McKenzie; first author on the resulting *Brain Sciences* paper (Aug. 2026).
 - **Work:** Screened PubMed and extracted data from the 60 included studies (192 arms; rabbit, rat, pig, dog, cat, human); co-developed the R analysis and dose-response figures; drafted and revised the manuscript through two rounds of review.
- **Zenith Legal** Washington, DC
Legal Recruiting Helper/Intern Aug. 2025 – Dec. 2025
 - **Focus:** Finding candidates for legal placement
 - **Candidate Sourcing:** Researched attorney candidates for firm placements through web scraping and database analysis. Wrote and memorized "sales pitch" for convincing candidates.
 - **Professional communication:** Made countless cold calls, emails to lawyers and law school students. Discussed recruiting via phone / zoom conversation. Attended and presented in daily meetings.
 - **Tech work:** Worked on maintaining app and website. Built CRM. Solo-built mobile app.
- **Krishnan Group Laboratory, University of Chicago** Chicago, Illinois
Summer Research Intern May 2024 – Aug. 2025
 - **Focus:** Research on ion transporters involved in neurodegenerative and inflammatory disease. Interned during summer of 2024 and 2025.
 - **DNA Nanotechnology:** Worked with the group's DNA nanodevices, including DNA-scaffolded quantum dot constructs, used as targeted fluorescent probes for measuring the chemistry of specific organelles inside living cells.
 - **Lab Work:** Maintained and performed experiments on *C. elegans* (RNAi KO and KD), maintained neural cell cultures, microscopy-based phenotypic assays, flow cytometry, patch clamp ephys, and general wetlab procedures under BSL-2 protocols.
 - **Scientific Practice:** Independently studied primary literature to support experimental work; maintained detailed lab notebooks documenting protocols, observations, and results; attended ≈ 15 seminars, professional talks, and thesis defenses.

PUBLICATIONS

- **Kalfus, J.,** Ward, D., Wróbel, B., & McKenzie, A. T. (2026). How Long Is the Brain Perfusable After Global Ischemia? A Systematic Review. *Brain Sciences*, 16(8), 882. [doi:10.3390/brainsci16080882](https://doi.org/10.3390/brainsci16080882)
- **Kalfus, J.,** & Lidický, B. (2026). An automated proof that $R(B_8, B_{10}) = 37$. *arXiv* preprint, math.CO. [arXiv:2606.05629](https://arxiv.org/abs/2606.05629). Found with AutoMath, my autonomous math-research harness; upper bound formalized in Lean.
- **Kalfus, J.,** & Goyal, S. (2026). BabblorLM: Targeting Closed-System Self-Generation at a 10M-Word Model's Own Reducible-Loss Spans. Under review, *BabyLM Challenge 2026* (EMNLP 2026 Workshop), OpenReview.

PROJECTS

- **Sports Betting Arbitrage Bot (2021):** Profited before permaban and legal threats. Built a bot that scraped offshore bookie lines (using BS/selenium, this was before the days of fast enough OCR), calculated arbitrage opportunities + Kelly calculation, also looked for EV bets, and automated betting on lines.
- **Tau.ai (2023/2024):** Led a team of high school students in building a CV-based eye tracker to help diagnose early Alzheimer's disease. Semifinalist in both the Conrad Challenge and Diamond Competition. See [project website](#) and [demo videos](#) for more information.
- **Bioinformatic Analysis Of PTSD and Alzheimer's (2024/2025):** Author of Research Paper analyzing common DEGs between PTSD and Alzheimer's Disease. Currently under review for publication.

- **Attempted Novel Alzheimer’s Treatment (2024/2025):** Theorized novel avenue for stopping neuronal degradation in Alzheimer’s Disease, screened 500+ compounds via molecular docking simulation against various receptors, created custom rating algorithm for determining best. Then tested the best candidate in-vitro, where it moderately improved cell viability. Unfortunately, later learned that the compound screened was under a class of patented Cyclophilin D inhibitors from about 5 years ago.
- **Synthesize (2025):** Chrome extension built to induce grapheme-color synesthesia by coloring text on websites
- **Gun Violence Survivors Aid (2022):** Website offering resources for victims of gun violence (no longer being maintained since 2023).

AWARDS

- **National Merit Scholarship Semifinalist:** Top 1% of high school seniors
- **AP Scholar with Distinction:** Highest AP-based award.
- **Wharton Investment Competition:** Led team of classmates in mock-investing challenge. Placed top 100 (monetarily) out of $\approx$ 5000 teams. Did NOT achieve semifinalist status.
- **ISEF Regeneron Biomedical Sciences Award:** Won \$375 for attempted Alzheimer’s treatment
- **HOSA Pharmacy Science International Finalist + Regional and State Champion:** Studied Pharmacology to compete in medical skills competition.
- **HOSA Pharmacy Science Mastery:** Scored >75% of all possible points on Pharmacy Science Exam
- **Conrad Challenge Innovator Award:** Won for “Tau.ai”
- **Global Seal of Biliteracy:** Working Fluency in Spanish

SKILLS

- **Software:** AutoDock Vina, GROMACS, Schrödinger, InCyte, Qlucore, Snapgene, AddGene, AutoCAD, Figma, Adobe Photoshop, Adobe Illustrator
- **Lab-based:** BSL-1/2 safety, cell culturing, *C. elegans* maintenance, flow cytometry, electrophysiology, ELISA, centrifugation, autoclaving, data analysis, report writing
- **Language Knowledge Matrix:**

	Programming	Human
Proficient	R (Bioconductor, EdgeR, cluster-Profiler), Python (NumPy, Pandas, Matplotlib, PyTorch)	English
Comfortable	Java, L ^A T _E X	Spanish
Decent	AutoHotKey, HTML, CSS	American Sign Language (ASL)

LICENSES AND CERTIFICATIONS

- **Pharmacy Technician License:** State of Alabama, Alabama Board of Pharmacy
- **Forklift Operator Certification:** United States, OSHA
- **Private Marine Vessel Operator:** United States, BoatUS

INDEPENDENT LEARNING

- **Johns Hopkins Center for Talented Youth (CTY):** Mathematics of Volatile Markets, Microeconomics
- **Student Institute for the Gifted (SIG):** Genetic Engineering, Astronomy
- **Centre for Effective Altruism:** Intro to Effective Altruism
- **Brown Pre-College:** Habitable Worlds: Possible Places for Life in the Solar System and Beyond
- **Khan Academy:** Numerous courses completed; 393,480 “Energy Points”